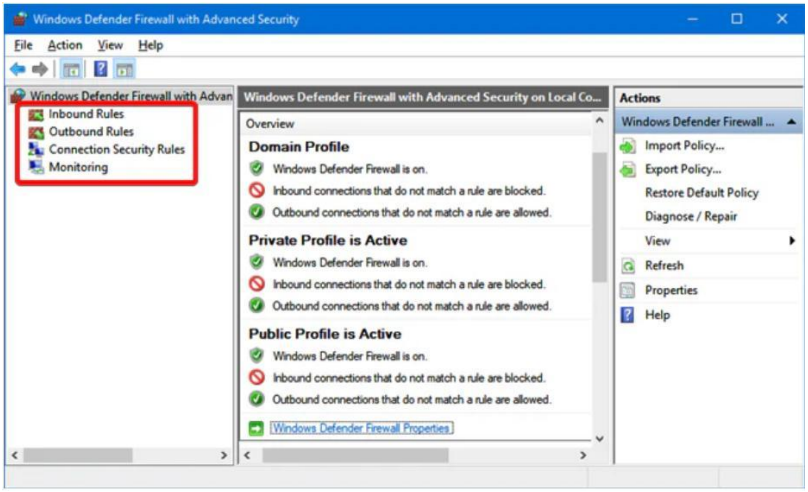
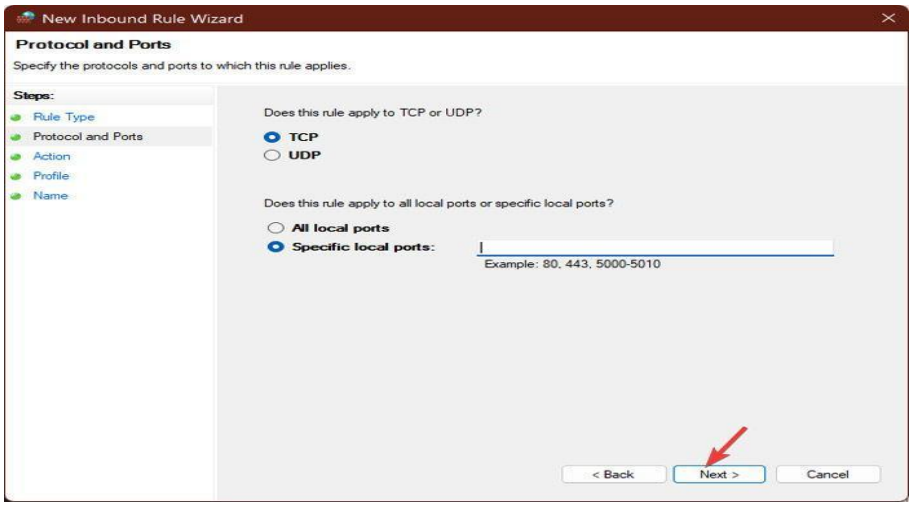
1) Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

Step 1: Open Windows Defender Firewall



1. Press Windows + R.
2. Type wf.msc.
3. Press Enter.
4. In the left panel, select Inbound Rules.

Step 2: Create a new inbound rule



1. Click New Rule… in the right-hand Actions panel.
2. Select Port, then click Next.
3. Select TCP.
4. Select Specific local ports and enter 8080.
5. Click Next.
6. Select Block the connection, then click Next.

7. Select all applicable profiles: Domain, Private, and Public.
8. Click Next.

Step 3: Name and save the rule

Enter the following details:

Rule name

Block Incoming Port 8080

Direction

Inbound

Protocol

TCP

Local port

8080

Action

Block the connection

Profiles

Domain, Private, Public

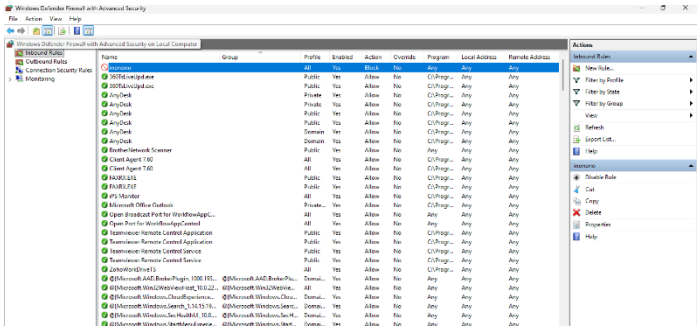
Click Finish to create the rule.

To block all incoming traffic using port 8080: Repeat the procedure for UDP, using the name Block Incoming Port 8080 UDP. The TCP rule alone does not block UDP traffic.

Step 4: Verify the rule

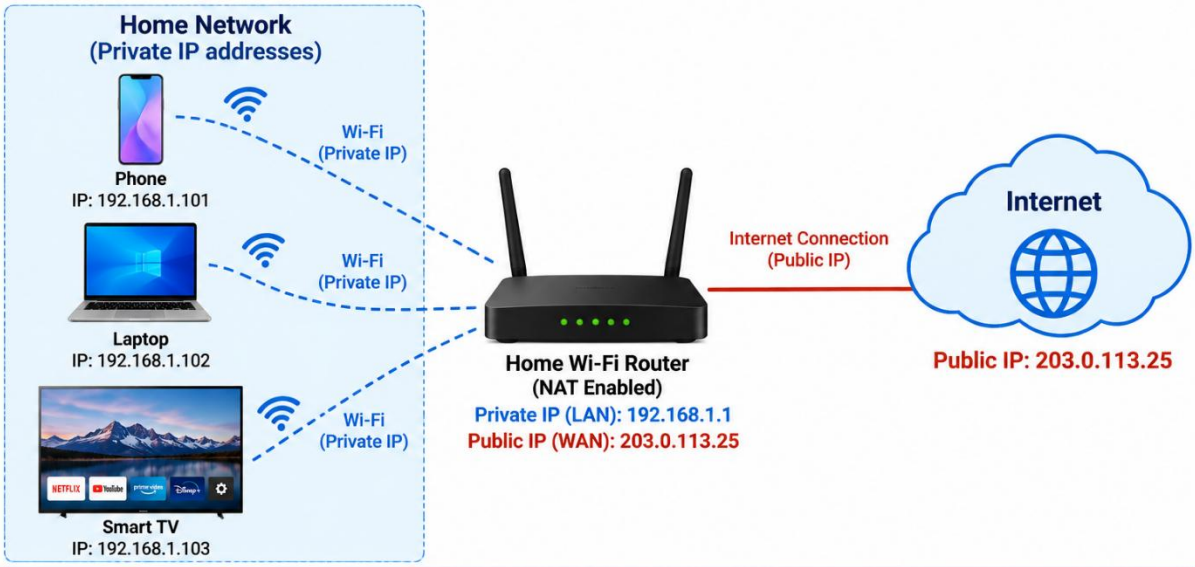
1. Return to Inbound Rules.
2. Find Block Incoming Port 8080.
3. Confirm that the rule is enabled and its Action is Block.
4. Open its Properties to verify the protocol, port and profiles.

Step 5: Take your screenshot



**2) Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.**



NAT Translation Table (Example)

Device	Private IP (Inside Network)	Source Port (Private)	Translated To (Public IP of Router)	Source Port (Public) (Assigned by NAT)
Phone	192.168.1.101	52345	203.0.113.25	40001
Laptop	192.168.1.102	52346	203.0.113.25	40002
Smart TV	192.168.1.103	52347	203.0.113.25	40003

3) Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Configure port filtering to block outgoing SMTP traffic on port 25

Objective: Configure the Wi-Fi router to block outgoing TCP traffic on port 25, which is commonly used for SMTP email transmission.

Step 1: Log in to the router

1. Connect your laptop to your Wi-Fi router.
2. Open a browser and enter the router's gateway address, such as 192.168.1.1.
3. Log in with your administrator credentials.

Step 2: Find the filtering settings

Look for a menu named Security, Firewall, Access Control, IP/Port Filtering, or Traffic Rules. The exact menu depends on your router model.

Step 3: Add an outbound filtering rule

If your router supports outbound port filtering, enter these example settings:

Setting	Value
Rule name	Block_SMTP_Port_25
Direction	Outbound
Protocol	TCP
Destination port	25
Action	Block / Deny
Status	Enabled

**4) Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.**

To determine whether the three packets are allowed or blocked, we need to compare their source IP addresses, destination IP addresses, and destination ports against the packet filtering rules provided by your trainer.

Since the rule set has not been included, the final decisions cannot be confirmed. Here is the packet analysis:

No.	Protocol	Source IP	Destination IP	Destination Port	Decision
1	HTTP (TCP)	192.168.1.5	172.217.166.78	80	Depends on rules
2	SSH (TCP)	10.0.0.10	34.201.23.1	22	Depends on rules
3	DNS (usually UDP)	192.168.1.7	8.8.8.8	53	Depends on rules

How to decide

1. HTTP request: Allow if the rules permit TCP traffic to destination port 80 and the destination IP is permitted; otherwise block.
2. SSH request: Allow if the rules permit TCP traffic to destination port 22 from source 10.0.0.10; otherwise block.
3. DNS query: Allow if the rules permit DNS traffic to destination port 53, usually UDP; otherwise block. Some DNS queries use TCP.